ELISA BRUNI

BIOGEOCHEMICAL MODELER

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WHO AM I

I hold a **PhD** in Soil Science.
I use mathematical models to understand the dynamics of **biogeochemical cycles** in terrestrial ecosystems. I am interested in **microbial controls** on soil organic matter turnover and their implications for sustainable land management and **climate mitigation**.

TECHNICAL SKILLS

Mathematical modeling of dynamical systems
Programming (R, python, Julia, Matlab, basic Fortran and Linux)

LANGUAGES

English
French
Italian
Spanish
Portuguese

WORKING EXPERIENCE

Visiting Fellow | FairCarboN mobility grant
Max Planck Institute for Biogeochemistry | Jena, Germany
March 2025

Integrating microbial diversity into soil carbon models: a hybrid modeling approach.

Post-doc | Sharing-MED
ENS (Ecole Normale Supérieure de Paris) | France
March 2024 - present

Studying the effect of conservation agriculture on soil health in Mediterranean drylands - modeling and data management platform. Development of a web tool to support decision-makers

Post-doc | Holisoils
ENS (Ecole Normale Supérieure de Paris) | France
April 2022 - February 2024

Multi-modeling approach to study the effect of forest management on soil organic carbon stocks and greenhouse gas emissions in Europe. Integrating microbial diversity in soil C and N models.

Research fellowship | FAPERJ
UFRRJ (Universidade Federal Rural de Rio de Janeiro) | Brazil
April 2022 - July 2022

Modeling the evolution of soil organic carbon stocks under conservation agriculture in tropical and subtropical climates

EDUCATION AND TRAINING

PhD | CLAND
LSCE (Laboratoire des Sciences du Climat et de l'Environnement) - Ecosys (INRAe) | France
December 2018 - March 2022

Soil organic carbon modeling: estimating carbon input changes required to reach policy objectives aimed at increasing soil organic carbon stocks - a multi-modeling approach

Training Course | ECOLAB | Cornell University, USA
June 2023 New advances in Land Carbon Cycle Modeling

Summer School | DySOM | TUM Institute Germany
September 2021 Dynamics of soil organic matter

Master II | CLUES
Climate, Land-Use and Ecosystem Services - AgroParisTech | France
2017 - 2018

Master I | EEET
Environmental, Energy and Transportation Economics - Université Paris Nanterre | France
2016- 2017

Bachelor Degree
Industrial Engineering - University of Bologna | Italy |
2012 - 2016

SELECTED PUBLICATIONS

Bruni, E., Yuste, J. C., Menichetti, L., Flores, O., Guasconi, D., Guenet, B., Hereș, A.-M., Lehtonen, A., Mäkipää, R., Pallandt, M., Pérez-Izquierdo, L., Richy, E., Santonja, M., Tupek, B., and Manzoni, S.: Microbial biomass – not diversity – drives soil carbon and nitrogen mineralization in Spanish holm oak ecosystems, Geoderma, 460, 117408, https://doi.org/10.1016/j.geoderma.2025.117408, 2025.

Bruni, E., Lugato, E., Chenu, C., and Guenet, B.: European croplands under climate change: Carbon input changes required to increase projected soil organic carbon stocks, Science of The Total Environment, 954, 176525, https://doi.org/10.1016/j.scitotenv.2024.176525, 2024.

Bruni, E., Guenet, B., Clivot, H., Kätterer, T., Martin, M., Virto, I., and Chenu, C.: Defining Quantitative Targets for Topsoil Organic Carbon Stock Increase in European Croplands: Case Studies With Exogenous Organic Matter Inputs, Front. Environ. Sci., 10, 824724, https://doi.org/10.3389/fenvs.2022.824724, 2022.

Bruni, E., Chenu, C., Abramoff, R. Z., Baldoni, G., Barkusky, D., Clivot, H., Huang, Y., Kätterer, T., Pikuła, D., Spiegel, H., Virto, I., and Guenet, B.: Multi-modelling predictions show high uncertainty of required carbon input changes to reach a 4‰ target, European J Soil Science, 73, https://doi.org/10.1111/ejss.13330, 2022.

Bruni, E., Guenet, B., Huang, Y., Clivot, H., Virto, I., Farina, R., Kätterer, T., Ciais, P., Martin, M., and Chenu, C.: Additional carbon inputs to reach a 4 per 1000 objective in Europe: feasibility and projected impacts of climate change based on Century simulations of long-term arable experiments, Biogeosciences, 18, 3981–4004, https://doi.org/10.5194/bg-18-3981-2021, 2021.

PROJECTS

Web interface development | me4soc

May 2024 v1 release - The **me4soc** platform for decision support:
http://me4soc.geologie.ens.fr/

SELECTED INTERNATIONAL CONFERENCES AND INVITED TALKS

Webinar UNFCCC – April 2024 – Soil Organic Carbon to Support National GHG Inventories: http://bit.ly/3UP0qfi

Centennial IUSS – 19-21 May 2024 – Florence (Italy) – ORAL – Conservation agriculture increases topsoil organic carbon stocks in Brazil – A multi-modeling approach.

SoilCET – 24-26 January 2024 – Rueil Malmaison (France) - ORAL - Improving the simulation of CO2 fluxes integrating microbial biodiversity into a simple soil organic carbon model.

EGU23 – 24–28 April 2023 – Vienna (Austria) - ORAL - Modelling soil organic carbon stocks and greenhouse gases in European forests with multi-model ensembles.

WCSS22 (World Congress of Soil Science) – 31 July - 5 August 2022 – Glasgow (UK) – ORAL - Reaching the 4 per 1000 target in Europe: feasibility and additional carbon input required

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ISMC (International Soil Modeling Consortium) – 18-22 May 2021 – ORAL - Increasing soil organic carbon stocks in agricultural fields: a multi-modeling analysis evaluating the carbon inputs required to maintain and increase soil organic carbon stocks in Europe.

Eurosoil – 23-27 August 2021 – Geneva (Switzerland) – ORAL - Increasing soil organic carbon storage: the 4 per 1000 objective through a modeling based approach.

TEACHING ACTIVITIES

2020-present | **Soil organic matter modeling** | M2 CLUES Module «Understanding and managing soil organic matter for ecosystem services delivery» AgroParisTech (France)

2024-2025 | **Introduction to process-based Modeling** | SAFEST Summer School - UniPisa (Italy)

2022-2023 | **Modeling soil organic carbon** | Departamento do solo UFRRJ (Brazil)

2020-2021; 2024 | **Numerical Modeling** | M2 CLUES Module «Numerical modelling» AgroParisTech (France)

2021 | **Modeling** | M2 GSSE «Gestion des sols et services écosystémiques» AgroParisTech (France)

2020-2021 | **Internship Tools** | M2 CLUES Module «Conseils et outils pour un bon stage» AgroParisTech (France)

2020 | **Bioclimatology** | 1A Module «Ecosystèmes continentaux» AgroParisTech (France)

2019-2020 | **Industrial Ecology** | 1A Module «Sciences de la production et la transformation» AgroParisTech (France)

STUDENT MENTORING

2025 M2 Prince Armand Koumi **Dzudzogbe** | Machine learning project to link soil health indicators to land management

2025 L3 Claire **Bodivid** | ORCHIDEE simulations of litter bag decomposition

2025 M2 Yanis **Hemeray** | Evaluation of soil organic carbon models performance in European forest soils

2025 M1 Illian **Brasselet-Darracq** | Development of the m4soc platform for the evaluation of soil carbon models

2025 L3 Dalia **Fages-Gouyou** | Evaluation of soil organic carbon models performance on Holisoils sites

2024 M2 Yasmine **Derbal** | Effects of land management changes on soil organic carbon stocks in Mediterranean drylands

2020 M2 Martin **Schwartz** | Restitution of crop residues as a strategy to increase soil organic carbon

AWARDS

2025 **International mobility grant FairCarboN (Jena, Germany)**
Integrating microbial diversity into soil carbon models: a hybrid modeling approach

2021 **International mobility grant FAPERJ-France (Rio de Janeiro, Brazil)**
Studying the effect of conservation agriculture in tropical and subtropical soils